CHAPTER C

Assessments and annual preparation

Versions T1 and T2 are available to pupils. Versions A and B and all the solutions are reserved for the teacher. The papers are original; they are not presented as official past papers. The approaches change between versions: direct or inverse model, optimisation or relative position, direct or reversed tree, projection or locus.

EV1-T1 · Common written test

Practice paper . Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 9 C. 8 D. 15

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 8 C. 2 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 60 B. 64 C. 96 D. 100

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. $\frac{9}{10}$ D. 1

7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. 6 D. -6

8. The slope between (0; 3) and (2; 9) is
   A. 2 B. -3 C. 6 D. 3

9. The mean of $3, 5, 7$ is
   A. 6 B. 15 C. 3 D. 5

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{3}{4}$ D. $\frac{1}{3}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 7 B. 5 C. 12 D. -1

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · A book collection (5 points)

A library has 120 books at the end of the reference month, called month 0. At the end of each following month, it adds 18 books to its collection; no books are removed. Let u_n the number of books available at the end of month n, for every natural number n.

1. Determine the type of (u_n), its common difference or ratio and its expression in terms of n. (1 pt)

2. Calculate u_8, then $u_0 + u_1 + \cdots + u_8$. Explain what this sum accumulates and why it does not count only distinct books. (2 pts)

3. Determine the first month in which the collection strictly exceeds 300 books. Check the previous month. (2 pts)

Exercise 2 · Area and shape choice (5 points)

A rectangle has a perimeter of 28 m. If one of its sides measures x metres, the other measures $14 - x$ metres. Its area is modelled by the function A defined on [0; 14] by $A(x) = x(14 - x)$, in square metres. The endpoints represent flattened rectangles.

1. Expand, then write A in vertex form. (1 pt)

2. Determine the maximum area and the corresponding value of x without using derivatives. (2 pts)

3. Solve $A(x) \geq 40$ on the domain. (2 pts)

Exercise 3 · A right angle (4 points)

The plane has an orthonormal frame. Consider the points $A(0; 1)$, $B(3; 2)$ and $C(-1; 4)$. The lengths are expressed in the unit of the coordinate system and areas in square units.

1. Calculate the coordinates of the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$. (1 pt)

2. Prove that these vectors are orthogonal and state the type of triangle. (1 pt)

3. Calculate the length BC then the area of triangle ABC. (2 pts)

EV1-T2 · Common written test

Practice paper . Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $4 + 2 \times 3$.
   A. 18 B. 9 C. 13 D. 10

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^4}{2^2}$.
   A. 2 B. 4 C. 1 D. 9

4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 10 D. 60

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 100 B. 96 C. 64 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. 1 B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. $\frac{9}{10}$

7. The solution of $2x + 4 = 0$ is
   A. 2 B. 8 C. -2 D. -8

8. The slope between (0; 4) and (2; 10) is
   A. 3 B. -3 C. 6 D. 2

9. The mean of $4, 6, 8$ is
   A. 18 B. 6 C. 7 D. 4

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 4$, $f(-2)$ is
    A. 6 B. 16 C. 8 D. 0

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Two growth processes (5 points)

We compare two models for a workshop's output. On day 0, each predicts 30 objects. Model A predicts 10 extra objects each day; model B predicts a doubling daily. The outputs on day n are therefore $a_n = 30 + 10n$ and $b_n = 30 \times 2^n$, for $n \in \mathbb{N}$. A cumulative total here means the sum of the outputs on the given days.

1. State the type of each sequence and calculate the production on day 2. (1 pt)

2. Calculate the cumulative production for days 0 to 3 in each model. (2 pts)

3. Check that model B exceeds model A on day 1, then explain why this gap remains positive on every following day by comparing the daily increases. (2 pts)

Exercise 2 · Imposed roots (5 points)

We are looking for a polynomial function f of degree 2, defined on $\mathbb{R}$. Its graph crosses the x-axis at the x-values -1 and 5, and it passes through the point (0; 10). This information should be enough to determine the function.

1. Give its factorised form. (1 pt)

2. Determine its sign, then solve $f(x) \leq 0$. (2 pts)

3. Give its vertex form and its maximum. (2 pts)

Exercise 3 · A controlled approximation (4 points)

The function f is defined on $\mathbb{R}$ by $f(x) = x^2 + 1$. We want to approximate $f(3{,}02)$ using the tangent to its curve at the point with x-coordinate 3. Calculations are done without a calculator.

1. For $h \neq 0$, calculate and simplify the rate of change between 3 and $3 + h$. (1 pt)

2. Deduce $f'(3)$, then the equation of the tangent at the point with x-coordinate 3. (1 pt)

3. Use the tangent to approximate $f(3{,}02)$. Then calculate the error "exact value minus approximation". (2 pts)

EV2-T1 · Mock exam 1

Practice paper . Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14-16 January 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $2 + 2 \times 3$.
   A. 8 B. 12 C. 11 D. 7

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{3}$ C. $\frac{2}{5}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 2 B. 7 C. 1 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 60 C. 70 D. 10

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 96 B. 100 C. 64 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. $\frac{9}{10}$ C. 1 D. $\frac{21}{20}$

7. The solution of $2x + 2 = 0$ is
   A. -4 B. -1 C. 1 D. 4

8. The slope between (0; 2) and (2; 8) is
   A. 6 B. 2 C. 3 D. -3

9. The mean of $2, 4, 6$ is
   A. 4 B. 2 C. 12 D. 5

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 4 B. 8 C. 6 D. -2

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Manufacturing and inspection (5 points)

Two workshops A and B produce 30 per cent and 70 per cent respectively of the items made by a company. Of the items from A, 10 per cent are defective; the proportion is 20 per cent for B. An item is chosen at random from the total production. The events A and D mean "comes from workshop A" and "is defective" respectively.

1. Construct a weighted tree whose first level distinguishes the two workshops. (1 pt)

2. Calculate $P(D)$, then the probability that a defective item comes from A. (2 pts)

3. Are the events A and D independent? Justify numerically. (2 pts)

Exercise 2 · Minimum of a quotient (5 points)

The function f is defined on the interval $]0; +\infty[$ by $f(x) = x + \dfrac{9}{x}$. We study its variations and certain tangents to its curve. The requested coordinates must be given in exact form.

1. Calculate and factorise the derivative. (1 pt)

2. Study the variation and give the minimum. (2 pts)

3. Determine the points where the tangent has slope $\dfrac{3}{4}$. (2 pts)

Exercise 3 · Projection and circle (4 points)

In an orthonormal coordinate system, consider the point $P(5; 2)$ and the line d with equation $x + y - 3 = 0$. Let H be the orthogonal projection of P on d.

1. Give a non-zero normal vector to d. (1 pt)

2. Determine the coordinates of H and check the membership and orthogonality conditions. (1 pt)

3. Give an equation of the circle with centre P through H. (2 pts)

EV2-T2 · Mock exam 1

Practice paper . Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14-16 January 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 8 C. 15 D. 9

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{1}{6}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 2 C. 8 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 60 D. 10

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 64 B. 96 C. 100 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{9}{10}$ B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. 1

7. The solution of $2x + 3 = 0$ is
   A. 6 B. -6 C. $\frac{3}{2}$ D. $\frac{-3}{2}$

8. The slope between (0; 3) and (2; 9) is
   A. -3 B. 2 C. 6 D. 3

9. The mean of $3, 5, 7$ is
   A. 3 B. 5 C. 6 D. 15

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 12 B. 7 C. -1 D. 5

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Two draws (5 points)

An urn contains four red balls and two blue balls, indistinguishable to the touch. We draw two balls at random, one after the other, without replacing the first. At each draw, the balls still in the urn are equally likely. The third question uses a different protocol, with replacement.

1. Calculate the probability of getting two red balls. (1 pt)

2. Calculate the probability of getting one ball of each colour, taking into account the two possible orders. (2 pts)

3. The first ball is now put back before the second draw. Calculate the probability of getting one ball of each colour and compare it with that of the protocol without replacement. (2 pts)

Exercise 2 · One box (5 points)

We cut out squares with side x cm from the four corners of a square sheet with side 10 cm. Folding up the edges forms an open box. Its volume, in cubic centimetres, is $V(x) = x(10 - 2x)^2$. This model is studied on [0; 5], including the flat endpoint cases.

1. Calculate V' in factorised form. (1 pt)

2. Study the variation. (2 pts)

3. Determine the exact maximum volume and justify that the endpoints do not give the maximum. (2 pts)

Exercise 3 · Identification and intersection (4 points)

In an orthonormal coordinate system, consider the set $\mathcal{C}$ of the points $M(x; y)$ which satisfy $x^2 + y^2 + 4x - 2y - 4 = 0$. Once it is identified, we will study its intersections with a line and a tangent.

1. Show that $\mathcal{C}$ is a circle and give its centre and radius. (1 pt)

2. Determine all points common to $\mathcal{C}$ and to the line $x = -2$. (1 pt)

3. Write the equation of the tangent to the circle at the point of intersection with the larger y-coordinate. Justify its direction. (2 pts)

EV3-T1 · Mock exam 2

Practice paper . Content of Chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1-3 April 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $6 + 2 \times 3$.
   A. 12 B. 24 C. 11 D. 15

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{5}{3}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 1 B. 11 C. 2 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 70 D. 60

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 64 B. 96 C. 100 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{21}{20}$ B. $\frac{9}{10}$ C. 1 D. $\frac{4}{5}$

7. The solution of $2x + 6 = 0$ is
   A. 12 B. -12 C. -3 D. 3

8. The slope between (0; 6) and (2; 12) is
   A. -3 B. 3 C. 2 D. 6

9. The mean of 6, 8, 10 is
   A. 9 B. 8 C. 24 D. 6

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{3}{5}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · A compound gain (5 points)

A game consists of two independent trials, with a probability of success of $\frac{1}{2}$ on each trial. Each success earns 3 monetary units; taking part costs 2 units. Let X be net gain, i.e. the total received minus the entry fee paid.

1. Determine the probability distribution of the net gain X. (1 pt)

2. Calculate its expectation and variance. (2 pts)

3. Determine the entry fee that would make the game fair. Explain what would happen to the standard deviation of the net gain. (2 pts)

Exercise 2 · Peak of a model (5 points)

A quantity is modelled for $t \geq 0$ by the function $f(t) = (t+2)e^{-t}$. The time t is expressed in a fixed unit. We seek its maximum value, then an approximation near the initial time.

1. Differentiate f. (1 pt)

2. Study the variation and give the maximum. (2 pts)

3. Write the tangent at 0 and approximate $f(0{,}1)$. (2 pts)

Exercise 3 · Unit circle (4 points)

The plane is equipped with an orthonormal coordinate system. On the unit circle with centre O, the points A and B are the respective images of the real numbers $\frac{\pi}{6}$ and $5\frac{\pi}{6}$.

1. Give their coordinates. (1 pt)

2. Calculate AB. (1 pt)

3. Give the equation of their perpendicular bisector and justify geometrically. (2 pts)

EV3-T2 · Mock exam 2

Practice paper . Content of Chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1-3 April 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $2 + 2 \times 3$.
   A. 11 B. 7 C. 12 D. 8

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 7 B. 4 C. 2 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 10 D. 60

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 96 B. 60 C. 100 D. 64

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$

7. The solution of $2x + 2 = 0$ is
   A. -1 B. 4 C. -4 D. 1

8. The slope between (0; 2) and (2; 8) is
   A. 3 B. 2 C. 6 D. -3

9. The mean of $2, 4, 6$ is
   A. 2 B. 4 C. 5 D. 12

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{1}{3}$ D. $\frac{3}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{3}{5}$ C. $\frac{1}{5}$ D. $\frac{2}{5}$

12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 6 B. -2 C. 4 D. 8

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Simulation and average (5 points)

The random variable X takes the values 0, 2 and 6 with the respective probabilities $\frac{1}{2}$, $\frac{1}{4}$ & $\frac{1}{4}$. We then simulate samples of 25 independent observations of this variable; let m be mean observed in a sample.

1. Calculate the expectation $E(X)$. (1 pt)

2. Calculate $V(X)$ and then the exact standard deviation $\sigma(X)$. (2 pts)

3. Write the interval of the values of m satisfying $|m - E(X)| \leq \frac{2\sigma(X)}{5}$. Explain why a simulation over a finite number of samples does not guarantee an exact coverage percentage. (2 pts)

Exercise 2 · Compare two models (5 points)

We define on $\mathbb{R}$ the functions $f(x) = e^{2x}$ & $g(x) = 1 + 2x$, then their difference $h = f - g$. The last question considers the values of f at whole-number times, from index 0.

1. Calculate the derivative of $h = f - g$. (1 pt)

2. Study h and compare the curves. (2 pts)

3. For $u_n = e^{2n}$, give the common ratio and the sum of the first four terms. (2 pts)

Exercise 3 · A translated circle (4 points)

In an orthonormal coordinate system, consider the circle $\mathcal{C}$ with equation $(x-1)^2 + (y+1)^2 = 4$ and the line d with equation $y = -1$. Let A and B be their intersection points, A being the one with smaller x-coordinate.

1. Give centre and radius. (1 pt)

2. Determine the intersections with the line. (1 pt)

3. For M on the circle, calculate $\overrightarrow{MA} \cdot \overrightarrow{MB}$ where A and B are these intersections; justify. (2 pts)

EV4-T1 · EAM preparation

Practice paper . The whole syllabus.

Duration: 2 h. No calculator. Marked out of 20. May and June practice; regional EAM to be confirmed.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $5 + 2 \times 3$.
   A. 14 B. 11 C. 21 D. 10

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{5}{6}$ C. $\frac{1}{6}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^5}{2^3}$.
   A. 1 B. 4 C. 10 D. 2

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 10 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 60 B. 96 C. 100 D. 64

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. 1 C. $\frac{9}{10}$ D. $\frac{21}{20}$

7. The solution of $2x + 5 = 0$ is
   A. $\frac{5}{2}$ B. 10 C. $\frac{-5}{2}$ D. -10

8. The slope between (0; 5) and (2; 11) is
   A. 3 B. -3 C. 6 D. 2

9. The mean of 5, 7, 9 is
   A. 7 B. 21 C. 5 D. 8

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$

12. For $f(x) = x^2 + 5$, $f(-2)$ is
    A. 20 B. 7 C. 1 D. 9

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Choosing a distribution (5 points)

We compare two games. The first gives a gain $X = 2$ with certainty. The second gives a gain $Y = -1$ with probability $\frac{2}{3}$ and a gain $Y = 8$ with probability $\frac{1}{3}$. Gains are expressed in the same currency unit; a negative value represents a loss.

1. Calculate $E(X)$ and $E(Y)$. (1 pt)

2. Calculate $V(Y)$ and compare the spreads of the two gains. (2 pts)

3. We play both games and add the winnings. Calculate $E(X+Y)$, citing the property used. (2 pts)

Exercise 2 · Discrete threshold and continuous model (5 points)

A quantity is modelled by $Q(t) = 64e^{-t}$ for $t \geq 0$. It is observed at integer times $n \in \mathbb{N}$ and we set $u_n = Q(n)$. For comparisons without a calculator, use the bounds $e^2 < 8 < e^3$.

1. Study the direction of variation. (1 pt)

2. Determine the first integer n such that $Q(n) < 8$ using the given bounds. (2 pts)

3. Give the type of $u_n = Q(n)$ and the expression of $u_0 + \cdots + u_3$. (2 pts)

Exercise 3 · Equation and circle (4 points)

In an orthonormal coordinate system, consider the set $\mathcal{C}$ defined by $x^2 + y^2 - 2x - 4y + 1 = 0$. The aim is to determine its characteristics, then to identify a particular tangent.

1. Show that $\mathcal{C}$ is a circle and state its centre and radius. (1 pt)

2. Determine its intersection points with the line with equation $x = 1$. (1 pt)

3. Determine the equation of the tangent to the circle at its highest point and justify. (2 pts)

EV4-T2 · EAM preparation

Practice paper . The whole syllabus.

Duration: 2 h. No calculator. Marked out of 20. May and June practice; regional EAM to be confirmed.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $6 + 2 \times 3$.
   A. 24 B. 15 C. 11 D. 12

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 2 B. 4 C. 1 D. 11

4. A value of 50 increases by 20%. Final value?
   A. 60 B. 10 C. 70 D. 40

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 64 B. 100 C. 96 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$

7. The solution of $2x + 6 = 0$ is
   A. -3 B. 12 C. 3 D. -12

8. The slope between (0; 6) and (2; 12) is
   A. 2 B. -3 C. 3 D. 6

9. The mean of 6 , 8, 10 is
   A. 6 B. 24 C. 8 D. 9

10. A sector of 90 ° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · A reversed tree (5 points)

Two events A and B satisfy $P(A) = \frac{3}{5}$, $P_A(B) = \frac{1}{3}$ & $P_{\overline{A}}(B) = \frac{3}{4}$. We want to construct a tree that starts by distinguishing B and $\overline{B}$, instead of A and $\overline{A}$.

1. Calculate $P(A \cap B)$ and $P(B)$. (1 pt)

2. Calculate $P_B(A)$ and $P_{\overline{B}}(A)$. (2 pts)

3. Construct and complete the weighted tree whose first level distinguishes B and $\overline{B}$, then the second A and $\overline{A}$. (2 pts)

Exercise 2 · Two optimisation methods (5 points)

The function f is defined on the interval [1; 6] by $f(x) = x(8 - x)$. We seek its extrema on this specific domain, then the values of x for which $f(x)$ reaches at least 15.

1. Give a vertex form and a maximum. (1 pt)

2. Find the maximum again by differentiation. (2 pts)

3. Determine the minimum on the domain and solve $f(x) \geq 15$. (2 pts)

Exercise 3 · Arguments and symmetry (4 points)

In an orthonormal coordinate system with origin O, the points A and B on the unit circle are the images of $\frac{7\pi}{6}$ and $11\frac{\pi}{6}$. The unsigned angle $\widehat{AOB}$ will be given in radians, between 0 and π.

1. Give their exact coordinates. (1 pt)

2. Calculate $\overrightarrow{OA} \cdot \overrightarrow{OB}$. (1 pt)

3. Determine the unsigned angle AOB, then AB. (2 pts)